Chapter 8 A test is a model comparison

Detailed notes pp. 1–3 and 14–15

Lecture slides PDF pp. 2–40

Name the population parameter before writing hypotheses. The **null hypothesis** H_0 specifies the reference claim; the **alternative** H_a specifies the departure the scientific question asks about. Choose its direction before inspecting the result.

$$\text{test statistic} = \frac{\text{estimate} - \text{null value}}{\text{standard error under the test model}}$$

A **null distribution** describes this statistic across repeated samples when H_0 is true. A large absolute value is meaningful only relative to the correct distribution and tail.

A p-value is conditional on the null model

The p-value is the probability, *assuming the null model*, of a statistic at least as extreme as the observed one in the direction(s) of H_a. It is not $P(H_0 \mid \text{data})$ or the probability the result is "due to chance."

For an observed statistic z with an appropriate standard-normal null reference,

Alternative	p-value	Reject at level α when
$\theta > \theta_0$	$1 - \Phi(z)$	$z > z_\alpha$
$\theta < \theta_0$	$\Phi(z)$	$z < -z_\alpha$
$\theta \neq \theta_0$	$2[1 - \Phi(\lvert z \rvert)]$	$\lvert z \rvert > z_{\alpha/2}$

Detailed notes pp. 14–15: critical values and rejection regions

Equivalent decision rule: reject when the p-value is at most the prespecified significance level α. Boundary equality has probability zero under a continuous reference distribution.

Write a conclusion about the parameter

- **Reject:** evidence against H_0 in favor of the stated alternative, at the chosen significance level and under the assumptions.
- **Fail to reject:** insufficient evidence against H_0; this does not prove it true or establish equality.

Small p-values do not measure effect size, scientific importance or causation. Report the observed estimate and context as well as the decision.

Keep in mind. A sampling design determines what can be inferred. A calculation cannot make a convenience sample representative or an observational comparison causal.

Chapter 8 Errors and power

Detailed notes pp. 16–19 and 25–27

Lecture slides PDF pp. 41–53; 74–82

Decision	H_0 true	Specified alternative true
Reject H_0	Type I error	Detection
Fail to reject	Correct non-rejection	Type II error

$$\alpha = P(\text{reject} \mid H_0), \quad \beta(\theta_1) = P(\text{fail to reject} \mid \theta = \theta_1), \quad \text{Power}(\theta_1) = 1 - \beta(\theta_1).$$

For an approximate test, the actual Type I probability is approximately the nominal α. Power depends on a specified true alternative, sample size, variability and rejection rule. There is no single β for every value in a composite alternative.

Compute power in two stages

1. Use the null distribution and α to establish the rejection region on the original estimate scale.
2. Keep that region fixed. Compute its probability under the specified alternative's sampling distribution.

For a two-sided mean test with constant SE s_0, set

$$L = \mu_0 - z_{\alpha/2}s_0, \qquad U = \mu_0 + z_{\alpha/2}s_0.$$

If $\bar{X}$ is approximately $N(\mu_1, s_1)$ under a specified alternative,

$$\beta(\mu_1) \approx \Phi\left(\frac{U - \mu_1}{s_1}\right) - \Phi\left(\frac{L - \mu_1}{s_1}\right).$$

For known, unchanged population SD, $s_1 = s_0 = \sigma/\sqrt{n}$. If variance changes with the parameter, the alternative SE must change too.

A left-sided proportion test illustrates that change

The null cutoff is $c = p_0 - z_\alpha\sqrt{p_0(1-p_0)/n}$. At true proportion p_1, the direct normal power approximation is

$$P_{p_1}(\hat{p} < c) \approx \Phi\left(\frac{c - p_1}{\sqrt{p_1(1-p_1)/n}}\right).$$

Larger n, smaller variability, or a larger effect in the tested direction generally improves power. Increasing α expands rejection and increases power but permits more Type I errors.

Keep in mind. Power is not 1 − p-value. Revisit the carpenter calculation, p. 18, and fitness calculation, p. 26, if the two-distribution distinction is unclear.

Chapter 8 Tests for means and paired data

Detailed notes pp. 20–22; one mean on 1–3 and 16

Lecture slides PDF pp. 54–65; one mean on 2–30

One population mean

Detailed notes pp. 2 and 16

For $H_0 : \mu = \mu_0$,

$$Z = \frac{\bar{X} - \mu_0}{\sigma/\sqrt{n}}.$$

With independent normal observations and known σ, this is exactly $N(0,1)$ under H_0. With an adequate iid finite-variance CLT, the normal reference is approximate. Chapter 8 may replace σ by s for a suitable large-sample approximation; that substitution is not an exact normal result.

Two independent population means

For $H_0 : \mu_1 - \mu_2 = \Delta_0$,

$$Z = \frac{(\bar{X}_1 - \bar{X}_2) - \Delta_0}{\sqrt{\sigma_1^2/n_1 + \sigma_2^2/n_2}}.$$

Use independent groups and adequate within-group sampling. Known SDs and normal populations give an exact normal reference. With unknown SDs and sufficiently large samples, replace σ_i by s_i for the course's approximate normal procedure. Unequal SDs are allowed in this large-sample formula; no pooled variance assumption is imposed.

The numerator order determines the sign of the estimate, alternative and conclusion. A negative observed difference favors a left-sided alternative, not automatically a two-sided rejection.

Paired observations are one sample of differences

For the same individuals measured twice, or explicitly matched pairs, define

$$D_i = X_i - Y_i, \qquad \mu_D = \mathrm{E}(D_i).$$

Analyze $D_1, \ldots, D_n$, where n is the number of pairs. The estimated SE is $s_D/\sqrt{n}$. A suitable large-sample normal test uses $(\bar{D} - d_0)/(s_D/\sqrt{n})$ approximately; Chapter 9 gives exact t_{n-1} inference for iid normal differences.

$$\mathrm{Var}(X - Y) = \mathrm{Var}(X) + \mathrm{Var}(Y) - 2\,\mathrm{Cov}(X, Y).$$

Pairing carries information about covariance. Do not replace s_D by $\sqrt{s_X^2 + s_Y^2}$ unless an appropriate covariance justification exists.

Keep in mind. The design decides paired versus independent. Equal sample sizes alone do not create pairs; the same individuals or meaningful matches do.

Chapter 8 Tests for proportions

Detailed notes pp. 23–29

Lecture slides PDF pp. 66–90

One proportion uses the null probability in its SE

For independent Bernoulli trials and $H_0 : p = p_0$, $\widehat{p} = x/n$ and

$$\boxed{Z = \frac{\widehat{p} - p_0}{\sqrt{p_0(1-p_0)/n}}.}$$

Check expected null successes $np_0 > 5$ and failures $n(1-p_0) > 5$, plus independence and an appropriate sampling fraction. This is a large-sample normal approximation. The Chapter 7 interval instead estimates SE with $\widehat{p}$; do not silently exchange the denominators.

Two proportions pool only under equality in the null

For independent groups and $H_0 : p_1 = p_2$, the null imposes a shared probability. Estimate that common value by the pooled proportion

$$\widehat{p}_{\text{pool}} = \frac{x_1 + x_2}{n_1 + n_2} = \frac{n_1\widehat{p}_1 + n_2\widehat{p}_2}{n_1 + n_2}.$$

It is weighted by sample size, not the simple average unless $n_1 = n_2$.

$$\boxed{Z = \frac{\widehat{p}_1 - \widehat{p}_2}{\sqrt{\widehat{p}_{\text{pool}}(1-\widehat{p}_{\text{pool}})(1/n_1 + 1/n_2)}}.}$$

Check the null expected success/failure counts in each group using the common pooled estimate. The observations and groups must be independent. The normal reference is approximate.

Do not transfer pooling to the confidence interval

A CI for an unknown difference uses the unpooled estimated SE

$$\sqrt{\frac{\widehat{p}_1(1-\widehat{p}_1)}{n_1} + \frac{\widehat{p}_2(1-\widehat{p}_2)}{n_2}}.$$

The two procedures ask different questions: the test evaluates an imposed equality; the CI estimates an unrestricted difference. Therefore a pooled test and an unpooled normal CI need not yield identical decisions at the same nominal level.

Carry the interpretation back to the population

Name the two population proportions, keep a fixed subtraction order and report a percentage-point difference. Statistical evidence for a difference does not itself show why it occurred.

Keep in mind. Do not use this equality-null pooled formula mechanically for a nonzero null difference. That is a different constrained problem.

Chapter 8 Choosing and checking a method

Detailed notes pp. 30–32

Lecture slides PDF pp. 2–90

Four questions before any calculation

1. **Target:** a mean, proportion, or difference? Name the population parameter.
2. **Design:** one sample, independent groups or paired observations?
3. **Reference:** why normal, and is that exact or approximate? Which SE is justified under the null?
4. **Direction:** left, right or two-sided, selected from the scientific question?

Confidence intervals and tests share a framework

For a two-sided normal test and matching interval that use the *same* SE and reference, rejecting $H_0 : \theta = \theta_0$ at level α is equivalent to excluding θ_0 from the $1 - \alpha$ CI. This equivalence requires matching procedures; the proportion formulas on p. 4 illustrate why it is not automatic.

Additional published connection: a Poisson mean

This connection appears on detailed p. 31; it is not a separate worked example in the Chapter 8 slides. For iid counts $X_i \sim \text{Poisson}(\lambda)$ with equal exposure,

$$\text{E}(X_i) = \text{Var}(X_i) = \lambda, \qquad \text{Var}(\bar{X}) = \lambda/n.$$

A sufficiently large-sample test of $H_0 : \lambda = \lambda_0 > 0$ can therefore use

$$Z = \frac{\bar{X} - \lambda_0}{\sqrt{\lambda_0/n}} \quad \text{with an approximate normal null reference.}$$

If calculating power at λ_1, retain the rejection rule and use the alternative SE $\sqrt{\lambda_1/n}$. Unequal exposure, dependence or variance exceeding the Poisson mean can invalidate this SE.

Use the existing practice in stages

The Friday questions begin on detailed p. 4; their released solutions follow later. The assigned two-proportion question is separate. Check the course page for released solutions rather than assuming an unreleased answer is available.

What Chapter 9 changes

Estimating variability from a small sample can change the reference distribution substantially. Under normal sampling, t inference accounts for that uncertainty. The estimate-minus-null structure and careful interpretation remain.

Keep in mind. Never turn “failed to reject” into “proved no effect,” or a small p-value into a measure of a large effect.